Journal of Functional Foods 21 (2016) 474–484

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Inhibitory effect of Erinacines A on the growth of DLD-1 colorectal cancer cells is induced by generation of reactive oxygen species and activation of p70S6K and p21

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ARTICLE INFO

Article history:
Received 3 July 2015
Received in revised form 12 December 2015
Accepted 23 December 2015
Available online

Keywords:
H. erinaceus
Erinacines A
Colorectal cancer cells
ROS
p70S6K
p21

ABSTRACT

Hericium erinaceus is a well-known edible mushroom with valuable biological properties. Erinacine A is the major active agent of the diterpenoid compounds of the cultured mycelia of *H. erinaceus*. This agent has multiple physiological activities, including anti-tumourigenic activity. In this study, we investigated the molecular mechanisms by which erinacine A mediated the generation of reactive oxygen species (ROS), and we performed cell cycle arrest to clarify molecular changes in colorectal cancer cells (CRC). Treatment of DLD-1 cells with erinacine A resulted in the phosphorylation of c-Jun N-terminal kinase (JNK) and p70S6K, the activation of p50 and p-IKB-β protein levels, the downregulation of cell-cycle-related proteins (cyclin A, cdk2, cyclin E, cdk4, and cyclin D1), and the induction of p21. Furthermore, treatment with the N-acetyl cysteine (NAC) and mTOR inhibitor (rapamycin) abolished erinacine A-induced cell cycle G1 arrest and reversed the association of CDK2 with Cyclin E and CDK4 with Cyclin D1. Therefore, when DLD-1 cells were grown as xenografts in nude mice, treatment with erinacine A induced a significant dose-dependent decrease in tumour growth. Histochemical and immunohistochemical analysis revealed that erinacine A treatment significantly reduced the number of mitotic cells. These results suggest that erinacine

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http://dx.doi.org/10.1016/j.jff.2015.12.031
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A presents an antitumour potential for CRC by inhibiting the growth of tumour cells *in vitro* and *in vivo*. Thus, we have identified a novel erinacine A-inhibited proliferation by activating p70S6K and ROS production, cell-cycle-related p21 proteins that provide a new mechanism for erinacine A-inhibited cell growth in human CRC.

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1. Introduction

Hericium erinaceus (Lion's mane or Yamabushitake) is an edible mushroom with medicinal properties that grows on old or dead broadleaf trees; it is used as a food in Japan and China without harmful effects. Both the mycelium and the fruiting bodies are the source of many bioactive extracts with chemopreventive efficacy. *H. erinaceus* has been extensively documented and possesses a range of therapeutic properties such as antioxidant activity (Malinowska, Krzyczkowski, Łapienis, & Herold, 2009), hypolipidaemic activity (Yang, Park, & Song, 2003), haemagglutinating activity (Gong et al., 2004), anti-microbial activity (Yim et al., 2007), and anti-ageing activity (Shimbo, Kawagishi, & Yokogoshi, 2005), immune modulation and anticancer activities (Lee & Hong, 2010; Li et al., 2014a, 2014b). More recent studies have revealed that erinacines (A–I), small molecular weight components that are the major active agent isolated from the cultured mycelia of *H. erinaceus*, are diterpenoid compounds that also play a role in varied functions, including neuroprotection through NGF modulation (Phan et al., 2014). Thus, *H. erinaceus* is attracting further attention as a novel resource for not only medicinal drugs, but also for dietary phytochemicals for disease prevention and health promotion through its biological properties (Jiang, Wang, Sun, & Zhang, 2014). It is important to note that the fruit body of *H. erinaceus* extractions seems to inhibit the extracellular signal-regulated kinase (ERK), c-Jun N-terminal kinase (JNK), and p38 mitogen-activated protein kinase (MAPK) phosphorylations. It also induces apoptosis, causes activity reduction of the matrix metalloproteinase (MMP) 2 and MMP-9 to block invasion of cancers, and shows anticancer efficacy against three gastrointestinal cancers: liver, colorectal and gastric in both *in vitro* cancer cell lines and *in vivo* tumour xenografts (Kim, Nam, & Friedman, 2013). Our previous investigation focused on exploring the biological agent of erinacine A from *H. erinaceus* mycelium, its structural elucidation by ethanol extraction, and HPLC analysis techniques (Lee et al., 2014). However, the mechanism by which *H. erinaceus* mycelium and its isolated diterpenoid derivative, erinacine A, initiates cell cycle arrest and how the signal cascades become activated remain poorly understood.

Colorectal cancer (CRC) is the third most common malignancy and the fourth leading cause of cancer death in the industrialised world (Haggar & Boushey, 2009). Proliferation of CRC cells usually fails to control checkpoints, one of which occurs in the mid-G1 phase (Sarsour, Kumar, Chaudhuri, Kalen, & Goswami, 2009). The inhibitors of the cell cycle progression stimulatory signalling pathway have been shown to induce the expression of Cdk inhibitors (CDKI p21, and p27), which bind to and subsequently inhibit cyclin-CDK activity, thereby blocking colorectal cells' proliferation and inducing cell growth arrest (Cheng et al., 1999; Graña & Reddy, 1995). Therefore, it is imperative to investigate how diet prevents the malignance and progression of CRC (Li-Weber, 2009). In addition to previous strategies that can reduce CRC risk, dietary phytochemicals are emerging as a promising chemoprevention approach. The cell cycle regulation and the *in vivo* anti-tumour effect of dietary *H. erinaceus* mycelium and its isolated diterpenoid derivative, erinacine A, are not clearly known in CRC.

Several reports support reactive oxygen species (ROS) functioning as second messengers regulating cellular processes. ROS participate in a wide variety of cellular functions, including cell proliferation, differentiation, and apoptosis (Valko et al., 2007). The role of intracellular ROS, the level of which is elevated and sensitive to oxidative damage, suggests that phenolic phytochemicals having antioxidant activity should short-circuit the signalling events and eventually inhibit cancer cell proliferation (Loo, 2003). However, the effect of erinacine A-mediated ROS production and the triggering of signalling pathways on human CRC remain unclear.

Chemopreventive compounds may inhibit cell proliferation through modulation of the MAPK pathways (Li-Weber, 2013). Major participants in this kinase cascade are 2 members of the MAPKa and mTOR/p70S6K pathways (Khanal et al., 2011; Teng, Kuo, & Sze, 2013). The essential role of the kinase cascade, as well as its mitochondrial-dependent ROS production pathway, during apoptosis induction by cellular stress and chemopreventive compounds, has recently been demonstrated. In our study, we first investigated the molecular mechanisms underlying erinacine A-induced ROS production and the activation of kinase cascades in human CRC. We also investigated the effect of erinacine A of *H. erinaceus* mycelium isolation on CRC cell cycle progression and *in vivo* antitumour potential. In the present study, we investigated how erinacine A inhibits CRC proliferation and cell cycle distribution by activating the mTOR/NFκB and p21 pathways and whether this activation is related to ROS generation. In addition, we found that erinacine A decreases the growth of CRC tumour cell xenograft in nude mice by inhibiting cell proliferation. Therefore, the development of more effective dietary phytochemicals for *H. erinaceus* mycelium is important.

2. Materials and methods

2.1. *Chemical reagents and antibodies*

All culture materials were purchased from Gibco (Grand Island, NY, USA). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), ROS scavenger (*N*-acetyl cysteine [NAC]), 2,7-dichlorodihydrofluorescein diacetate (H_2DCFDA), dihydroethidium (DHE), ERK inhibitor (PD98059), Janus kinase (JNK) inhibitor (SP600125), p38 inhibitor (SB203580), and mTOR inhibitor (rapamycin) were purchased from Sigma (St. Louis, MO, USA).

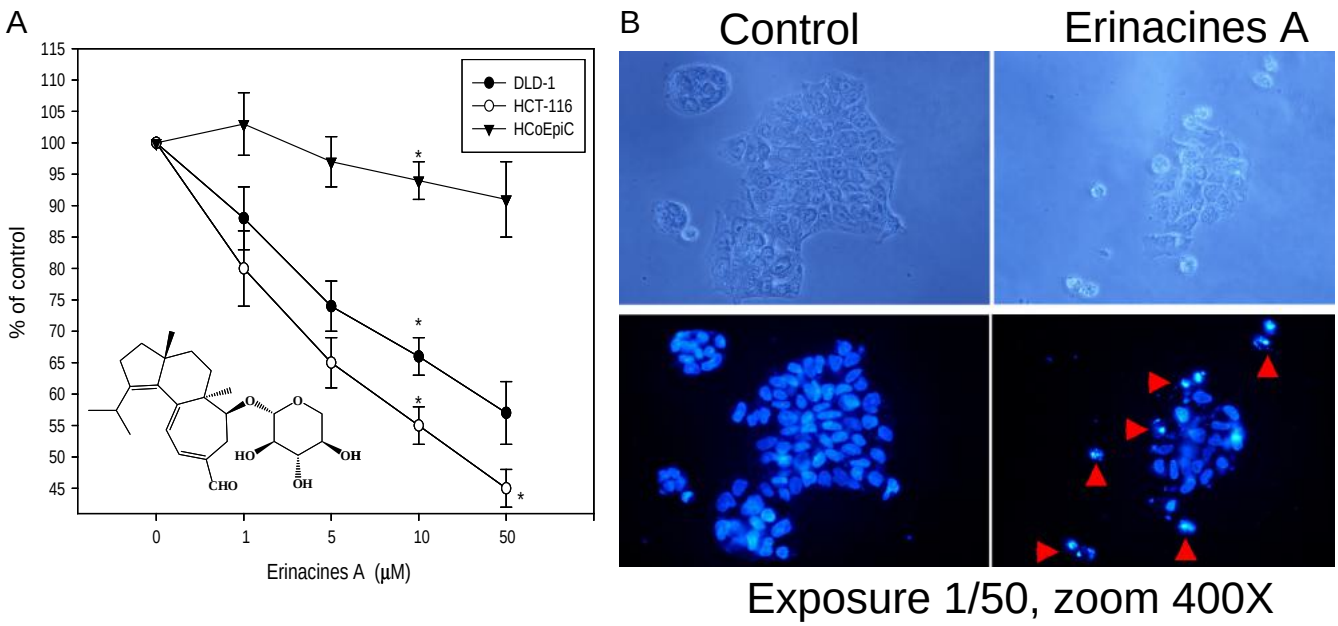


Fig. 1 – Effect of Erinacines A on cell viability and morphological characteristics of human colorectal cancer DLD-1 cells, and its role in assessing cell death. (A) DLD-1, HAT-116 and HCoEpiC cell were treated with either 0.1% DMSO (as control) or erinacine A (1–50 mM) for 24 h and the proportion of surviving cells was measured by the MTT assay. (B) Changes in nuclei by DAPI staining. DLD-1 cells were treated with vehicle or 50 mM erinacine A for 24 h, and stained with DAPI. Apoptotic cells were measured under fluorescence microscopy. The data were presented as the mean of three repeats from one independent experiment. Other data in this figure are presented as mean ± SD of three independent experiments. *Indicate means that are significantly different when compared to the control group of DLD-1 with P < 0.05.

Mouse monoclonal antibodies against cyclin A, cyclin D1, cyclin E, cyclin B, and β-actin were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). Rabbit polyclonal antibodies against cdk2, ERK1/2Thr^{202}Ty^{r204}, p38 Thr^{180}Tyr^{182}, and JNK1/2 Thr^{183}Tyr^{185} mouse monoclonal cdk1 antibody were purchased from Cell Signaling Technology (Beverly, MA, USA). The TdT-mediated dUTP Nick End Labeling (TUNEL) kits were from Roche. SDS, NP-40, sodium deoxycholate, protease inhibitor cocktails were purchased from Sigma.

2.2. H. erinaceus *extracts and analysis of erinacine A*

Fresh mycelium of *H. erinaceus* was extracted with ethanol. The extract was concentrated and fractionated by solvent partition between ethylacetate and water. The ethylacetate fraction was subjected to silica gel column chromatography using *n*-hexane–ethylacetate as the eluent. The hexane–acetone eluate was subjected to silica gel column chromatography according to previous studies (Lee et al., 2014; Li et al., 2014a, 2014b); HPLC analysis of erinacine A was executed according to the previous study with minor modifications. The analytical column used was a COSMOSIL 5C18-AR-II (250 × 4.6 mm; particle size 5 μm, Nacalai USA, Inc., Kyoto, Japan). Separation was performed at 40 °C using two different gradients for the mobile phase, which consisted of two solvents, methanol (A) and 2.0% acetic acid in water (B). The gradient elution had the following profile: 0–20 min, 60–90% (A); 20–25 min, 90% (A). The retention time of erinacine A was approximately ~17 min at a flow rate of 1.0 mL/min with a scanning UV wavelength at 340 nm. The 3 mg/g erinacine A in the *H. erinaceus* extracted with 85% ethanol was confirmed and quantified by HPLC as shown in Fig. 1 (Lee et al., 2014). Chemical compounds studied in this article Erinacine A was recorded (PubChem CID: 10410568).

2.3. *Cell culture*

Human colon cancer cell line DLD-1 (CCL-221) and human colorectal carcinoma cell line HCT-116 (CCL-247) were purchased from American Type Culture Collection (ATCC). DLD-1 cells were cultured in RPMI 1640 medium composed of 10% fetal calf serum (FCS) (S0113; Biochrom KG, Berlin, Germany) and 1% antibiotics (100 units/mL of penicillin and 100 μg/mL of streptomycin) (Sigma Chemicals, St. Louis, MO, USA) and incubated at 37 °C with 5% CO2. Adhered cells were washed twice with PBS. HCT-116 was cultured in DMEM supplemented with 10% heat-inactivated newborn calf serum at 37 °C in a humidified 5% CO2 incubator (Huang et al., 2012a, 2012b).

2.4. *Cell growth and proliferation assay*

The previously reported MTT quantitative colorimetric assay was verified as being capable of detecting viable cells and was used for cell viability determinations as previously described (Chiu et al., 2011). Cells were seeded and incubated with the various agents. Thereafter, the medium was changed, and cells were incubated with MTT (0.5 mg/mL) for 4 h. The viable cell number was directly proportional to the production of formazan, which was measured spectrophotometrically (λ = 563 nm) after solubilisation with isopropanol. Cell growth

was determined by counting the cells at the indicated time points with a Coulter counter, combined with a trypan blue (0.2%) exclusion assay.

2.5. *Apoptosis assay and cell cycle distribution analysis*

Changes in cell morphological characteristics during apoptosis were examined using fluorescence microscopy of 4′,6-diamidino-2-phenylindole (DAPI)-stained cells. The monolayer of cells was fixed with 4% paraformaldehyde for 30 min at room temperature. The fixed cells were permeabilised with 3 treatments in 0.2% Triton X-100 in phosphate-buffered saline, followed by incubation with 1 μg/mL of DAPI for 30 min. The apoptotic nuclei were detected under 200 × magnification using a fluorescent microscope with a 340/380 nm excitation filter and were scored according to the percentage of apoptotic nuclei found in samples containing 200 to 300 cells. Cell-cycle distribution was analysed using flow cytometry. Cells stained with propidium iodide were analysed with a FACScalibur™ (Becton Dickinson), and the data were analysed using a mod-fit cell cycle analysis program (Teng, Kuo, Cheng, Wang, & Sze, 2012).

2.6. *Preparation of total cell extracts and immunoblot analyses*

Cells were lysed with a buffer containing 1% NP-40, 0.5% sodium deoxycholate, 0.1% sodium dodecyl sulfate (SDS), and a protease inhibitor mixture (phenylmethylsulfonyl fluoride, aprotinin, and sodium orthovanadate). The total cell lysate (50 μg of protein) was separated by SDS-polyacrylamide gel electrophoresis (PAGE) (12% running, 4% stacking) and analysed by using the designated antibodies and the Western-Light chemiluminescent detection system (Bio-Rad, Hercules, CA, USA), as previously described (Chiu et al., 2011).

2.7. *Measurement of ROS*

The intracellular accumulation of ROS was determined by using the fluorescent probe H_2DCFDA, which is commonly used to measure H_2O_2, but is now accepted as also being sensitive to other peroxides. The oxidation-sensitive dyes DHE (2 μM) or H_2DCFDA (2 μM) were added separately to all tubes 15 min prior to harvest. Incubation was terminated by 10-fold dilution with ice-cold fluorescence-activated cell sorting (FACS) buffer, and the cells were washed prior to FACS analysis. ROS (O_2^-) generation was determined through increases in DHE. The fluorescence was measured at the desired time intervals using flow cytometry (Chiu et al., 2011).

2.8. In vivo *treatment*

BALB/c-*nu* mice were purchased from the National Laboratory Animal Center in Taiwan. BALB/c-*nu* female nude mice, 4–6 weeks old (18–20 g), were maintained under Specific Pathogen Free (SPF) conditions and supplied with sterilised food and water. After trypsinising, the DLD-1 cells (10^6 cells/0.2 ml) were injected subcutaneously into the flanks of female athymic BALB/c-*nu* mice (4–6 weeks old). After tumour inoculation, the mice were divided randomly into four groups of eight mice each. The control animals were treated daily with 0.1 mL DMSO (0.25%; i.p.), and the test animals were treated with erinacine A (1, 2, 5 mg/day; i.p.) for five days. Tumour volumes were monitored on the fourth day, and then measured at four-day intervals using calipers; calculation was based on the following formula: length × width^2 × π/6 (Kuo et al., 2006). To monitor drug toxicity levels, the body weights of the mice were measured every week. After 18 days, the mice were euthanised, and the tumours were removed and assayed. In addition, a pathologist examined the organs of each mouse, including the liver, lungs, and kidneys.

2.9. *Histochemistry and immunohistochemistry*

Tumour tissue sections were fixed in formaldehyde and embedded in paraffin blocks. The slides were stained with haematoxylin and eosin, and mounted for microscopic examination. For immunohistochemical analysis, 5 μm thick sections from each subcutaneous tumour specimen were fixed and incubated with monoclonal anti-p21 and p70S6K antibodies (Santa Cruz, CA, USA) and then with 1:100 diluted biotinylated horse anti-mouse IgG for 1 h. After being washing with PBS, they were reacted with 1:100 diluted avidin-biotin peroxidase mixture (Vectastain Universal Elite ABC Kit) for 30 min. The sections were washed thoroughly in PBS, and slides were counterstained with haematoxylin. Finally, the slides were washed, dehydrated, and mounted for microscopic examination. The immunoreative cells (brown) were counted. The digital images were captured using a digital camera (Canon A640). The positive area and optical density (OD) of positive cells + (brown) were determined by measuring three randomly selected microscopic fields (400 × magnification) for each slide. The IHC index was defined as average integral optical density (AIOD) (AIOD = positive area × OD/total area) (Chen, Kuo, Lee, & Tsai, 2014).

2.10. *Statistical analysis*

The experiments were performed in triplicate and data were reported as the mean ± standard deviation from 3 independent experiments and evaluated by one-way ANOVA. Significant differences were established at $P < 0.05$ (Chen et al., 2014).

3. Results and discussion

3.1. *Effects of erinacine A on the viability of human CRC cells*

In our previous study, the yield of 3 mg/g erinacine A in the fresh mycelium of *H. erinaceus* extracted with ethanol was confirmed and quantified by HPLC as shown (Lee et al., 2014). To determine whether isolated *H. erinaceus* mycelium, erinacine A, is cytotoxic to human CRC cells, we exposed human DLD-1, HCT-116 and normal human Colonic Epithelial Cells (HCoEpiC) to a range of erinacine A doses for 24 h and examined them by MTT assays. Erinacine A treatment resulted in a dose-dependent loss of cell viability, as shown in Fig. 1A. After treatment with 10 μM erinacine A for 24 h, 64 and 55% of DLD-1 and HCT-116 cells ($P < 0.01$), respectively, survived in culture

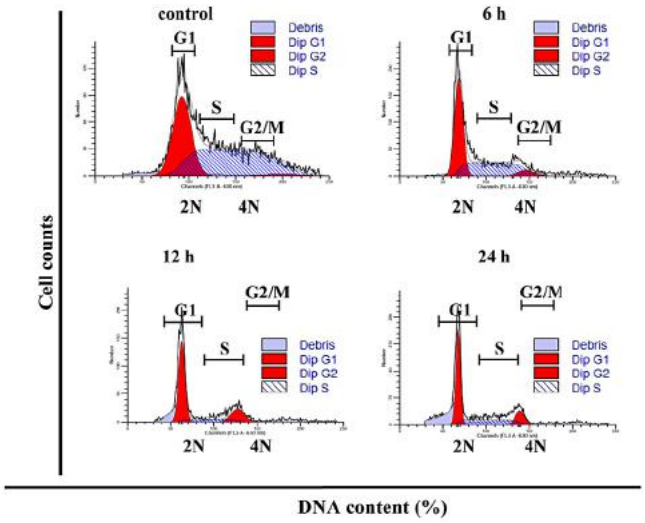
(Fig. 1A). However, erinacine A did not show significant cytotoxic effects in HCoEpiC cells. Furthermore, to verify CIL-102-induced cell toxicity, we also examined the changes in cell morphology after erinacine A exposure. Fig. 1B shows that exposure to erinacine A for 24 h causes DLD-1 cells to show characteristic features of cell shrinking, rounding, and becoming smaller growth parts.

3.2. *Erinacine A induces cell cycle G1 arrest in DLD-1 CRC cells*

Cell cycle distribution analysis showed that erinacine A induces cell cycle G1 arrest in DLD-1 cells. As demonstrated in Fig. 2, following a 24-h treatment with 10 μM erinacine A, the percentages of G1 phase cells increased to 54 ± 4 (6 h), 68 ± 2 (12 h), and 75 ± 2% (24 h), respectively. These results suggest that erinacine A induces G1 arrest in a time-dependent manner.

3.3. *Involvement of ROS accumulation in erinacine A-inhibited cell proliferation*

Recent theories pertain to the persistent oxidative stress of ROS formation in the inhibition of cancer cell proliferation by phytochemicals, which was concluded to be the potentiation of the Ras/MEK/MAPKs and PI3K/Akt/mTOR pathways (Khanal et al., 2011; Loo, 2003; Teng et al., 2013). In light of these theories, we determined the possibility that levels of ROS might mediate erinacine A-inhibited proliferation in this study model. To elucidate the events of erinacine A-inhibited U87 cell proliferation, we examined kinetic studies to evaluate intracellular ROS production by FACS. The kinetics of erinacine A-stimulated ROS production were measured using the cell-infusing oxidation-sensitive dye H_2DCFDA during a 24 h period. As shown in Fig. 3, erinacine A stimulated ROS production as early as 6 h, but it caused a 20-fold increase at 24 h, indicating the formation of oxidants in DLD-1 cells. Using H_2DCFDA as a specific probe for hydrogen peroxide (H_2O_2), we observed erinacine A-induced oxidation of H_2DCFDA. On the basis of these data, the 6-h time point was used in subsequent experiments for determining the concentration response of erinacine A stimulation of ROS generation.



	G1	S	G2/M
control	41±2	48±2	11±2
6 h	54±2	38±3	8±3
12 h	68±3	22±3	10±2
24 h	75±2	14±3	11±2

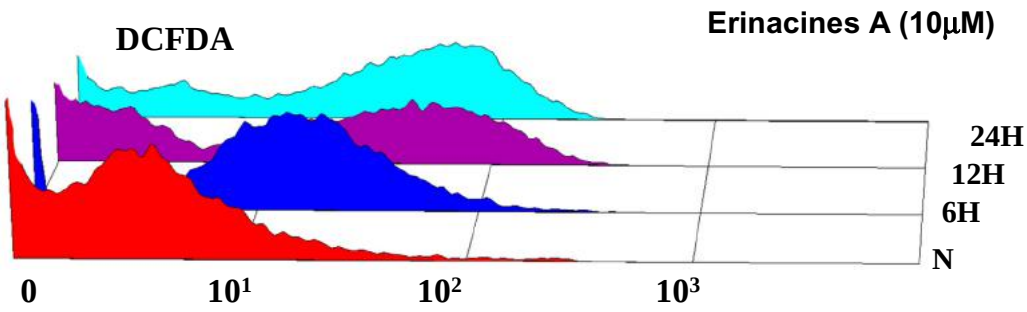
Fig. 2 – Effect of Erinacine A on cell cycle distribution in DLD-1 cells. (A) After treatment with Erinacine A (10 mM) for 24 h, the cells were fixed and stained with propidium iodide, and the DNA content was analysed by flow cytometry (FACS). The cell number percentage in each phase (G1, S, and G2/M) of cell cycle was calculated and expressed. Data are presented as means ± SD based on three independent experiments.

3.4. *Activation of JNK1/2 MAPKs and mTOR/p70S6K signalling pathways by erinacine A in DLD-1 cells*

Some dietary phytochemicals suppress carcinoma cells by inducing apoptosis. Numerous investigations have shown the oxidative stress response to cancer cells that initiates activation of the MAPKs/NFκB and mTOR/p70S6K signalling pathways (Alexander et al., 2010; Ray, Huang, & Tsuji, 2012). To demonstrate the effect of erinacine A on the activation of the kinase signalling pathway, we assessed whole-cell lysates from erinacine A-treated DLD-1 cells by Western blot analysis using antibodies against the activated phosphorylated forms of ERK1/2Thr^{202}Ty^{r204}, p38 Thr^{180}Tyr^{182}, and JNK1/2 Thr^{183}Tyr.^{185} As shown in Fig. 4A, treatment of DLD-1 cells with erinacine A resulted in a time-dependent phosphorylation of JNK1/2 and p70S6K over a 6-h period. Activation of both kinases was observed after treatment with erinacine A and significantly persisted for at least 6 h. ERK1/2 and p38 MAPK were found in DLD-1 cells but exhibited significantly inhibition after erinacine A treatment. To investigate whether NF-κB binds the promoter region in DLD-1, we performed quantitative analysis for NF-κB p50 binding activity *in vitro* by using TF ELISA kits from Panomics (Huang et al., 2012a, 2012b). The results from those experiments showed that treatment of erinacine A resulted in increased NF-κB binding activities (Fig. 4B).

3.5. *Effects of erinacine A on the expression of NF-κB, CDKIs and cell-cycle-related proteins in DLD-1cells*

Western blot analysis was used to determine whether erinacine A induces G1 cell cycle arrest by modulating the expression of cell cycle-related proteins, as shown in Fig. 5. The analysis showed a marked downregulation of cyclin A, CDK2, CDK4 as well as the association of CDK2/cyclin E and CDK4/cyclin D1 after erinacine A treatment of DLD-1 cells for a period of time points. In addition, it was found that levels of these proteins, NF-κB p50, p21 and p27, significantly increased compared with those in the control. The data clearly demonstrated that erinacine A-inhibited cell growth was associated with light production of ROS-dependent signal transduction and increases in the expression of p21 (due to NF-κB p50 binding activity).



	Fluorescence Intensity	Fold of Control
Control	4±2	1
6H-EA	22±4	5
12H-EA	74±3	18
24H-EA	82±3	20

Fig. 3 – Effect of Erinacine A on generation of ROS in DLD-1 cells. (A) Cells were treated with erinacine A for indicated times, and intracellular ROS were determined by FACS analysis as described in "Materials and methods." Representative histograms show typical FACS profiles for H2DCFDA in untreated (red peaks) or erinacine A-treated DLD-1 cells. The production of ROS was expressed as the percentage of cells in contrast with the untreated cells. Data are presented as means ± SD based on three independent experiments.

3.6. *ROS and JNK1/2 MAPKs and mTOR pathways are involved in the regulation of erinacine A–induced cell cycle G1 arrest in DLD-1 cells*

The sustained oxidative stress of ROS production by various kinds of signalling stimulation has been reported to play a role in cell death and survival (Valko et al., 2007). To investigate the roles of the MAPKs and mTOR signalling pathways and the role of ROS in erinacine A-induced G1 phase cell-cycle arrest, we exposed DLD-1 cells to erinacine A and then cotreated them with the specific ERK1/2 inhibitor PD98059, JNK inhibitor SP600125, mTOR inhibitor rapamycin, or the reactive oxygen intermediate scavenger NAC. The effects of those inhibitors in reversing the erinacine A-induced generation of ROS and the cell cycle distribution were determined. As shown in Table 1, mTOR inhibitor and NAC almost completely blocked erinacine A-induced cell cycle G1 arrest in DLD-1 cells. However, the inhibitors SB203580 and SP600125 had no effect on the reversion of cell cycle arrest at the G1 phase. We subsequently examined the putative role of ROS in erinacine A treatment in CRC cells. As shown in Table 2, mTOR inhibitor and NAC markedly reversed the erinacine A-induced level of ROS. Taken together, NAC significantly inhibited erinacine A induced cell cycle G1 arrest and production of ROS than mTOR inhibitor and SP600125 in DLD-1 cells.

To then determine the regulation of JNK1/2 MAPKs and mTOR pathways in constitutive amounts of ROS-mediated DLD-1 cell arrest, cells were incubated for 2 h with the following specific inhibitors: JNK inhibitor SP600125, ERK (PD98059), ROS scavenger (NAC), mTOR (rapamycin) and NF-κB p50 inhibitor (PDTC). They were then treated with erinacine A for 12 h. The expressions of NF-κB p50, p-IKBβ, p21 as well as the association of CDK2/cyclin E and CDK4/cyclin D1 were analysed. NAC, rapamycin and PDTC significantly reversed the erinacine A-induced expression of NF-κB p50, p-IKBβ, p21 and then reduced the erinacine A-induced expression of the association of CDK2/cyclin E and CDK4/cyclin D1. In contrast, SP600125 and PD98059 had no effects on erinacine A treatment (Fig. 6). Taken together, these results show that erinacine A induces sustained activation of mTOR/p70S6K/ NF-κB and p21, and that cellular oxidants (ROS) are required for erinacine A inhibition of DLD-1 cell growth.

Table 1 – Effects of the kinase inhibitor blocking Erinacines A-induced cell cycle arrest.

	G1	S	G2/M
Control	40 ± 2	47 ± 2	13 ± 2
EA	78 ± 3	12 ± 2	10 ± 2
EA-NAC	52 ± 3	32 ± 2	16 ± 2
EA-Rapamycin	58 ± 2	27 ± 2	15 ± 3
EA-SB	85 ± 5	26 ± 2	12 ± 3
EA-SP	70 ± 5	15 ± 3	11 ± 2

Table 2 – Effects of the kinase inhibitor blocking Erinacines A-induced generation of reactive oxygen species.

	Fluorescence intensity	Fold of control
Control	5 ± 2	1
EA	90 ± 3	18
EA-SP	38 ± 4	7
EA-NAC	22 ± 2	4
EA-Rapamycin	35 ± 4	7

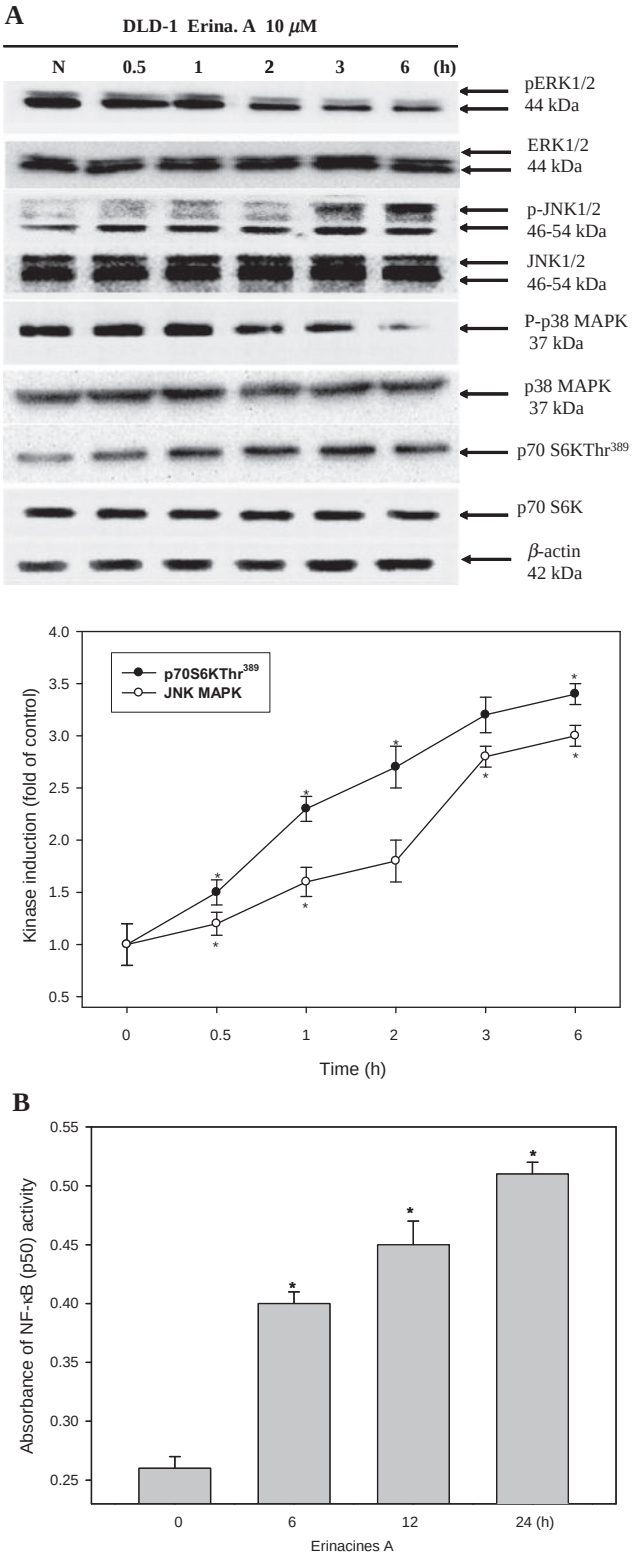


Fig. 4 – The expression levels of NF-κB-related proteins as well as phosphorylation and protein levels of ERK, JNK, p38 MAPKs and p70S6K after treatment with erinacine A. (A) DLD-1 cells were treated with 10 mM erinacine A for the indicated time points. Total cell lysates treated with erinacine A were analysed by SDS-PAGE and subsequently immunoblotted with antisera against phosphorylated ERK1/2, JNK, p38 and p70S6K as well as non-phosphorylated ERK1/2, JNK, p38 and p70S6K and β-actin, serving as an internal control. Protein levels were quantified by densitometric analysis with the ratio of untreated control being set 1 fold. The quantitative data were presented as the mean of three repeats from independent experiments. The data were presented as mean ± SD of three independent experiments. (B) NF-κB p50 activation was determined by a TF ELISA assay. All bar graphs represent absorbance, mean ± SD on three repeats from independent experiments. *P < 0.05, as compared to control group.

3.7. In vivo *inhibition of DLD-1 tumour growth by erinacine A*

To evaluate the *in vivo* effects of erinacine A on tumour growth, DLD-1 cells were xenografted into nude mice as described previously. The results of our previous toxicity-assessment experiments suggested that the *in vivo* dosage of erinacine A should not exceed 300 mg/kg/day from fractionation from *H. erinaceus* mycelium and erinacine A 5 mg/day (data not shown); therefore, daily intraperitoneal injections of erinacine A concentrations of 1, 2, and 5 mg were used in these experiment. Fig. 7A shows the time course of DLD-1 xenograft growth with and without erinacine A treatment. Evaluation of the xenograft tumour volume showed that a significant inhibition of tumour growth by erinacine A was caused in both a dose- and time-dependent manner. In the erinacine A-treated mice at 18 days after cell implantation, the DLD-1 xenograft volumes were inhibited to 66, 51, and 42% ($P < 0.05$), respectively, of those in the control group. No signs of toxicity were observed in any of the transplanted nude mice as assessed by body weight and microscopic examination of individual organs (data not shown). At the end of this experiment, the xenograft tumours were removed. Evaluation of a proliferative mitosis by H&E staining of tumour sections showed that erinacine A led to a decrease in the number of proliferating cells in all tumours compared with those untreated in the control group. In addition, another marker of the erinacine A target proteins p21 and p70S6K was examined by immunohistochemistry. The results showed that erinacine A treatment induced the number of p21 and p70S6K-positive cells. TUNEL staining was used to evaluate apoptosis in xenografts. Quantification of TUNEL staining showed that erinacine A treatment caused significantly more apoptosis in tumours derived than in the untreated group ($P < 0.05$; Fig. 7B). Fig. 7 shows representative images from all tumour sections analysed, in addition to photographs of s.c. tumours. These results indicate the antitumour action of erinacine A on DLD-1 cells *in vivo*.

Our *in vitro* data demonstrated that *H. erinaceus* mycelium contains erinacine A and treatment above a concentration of

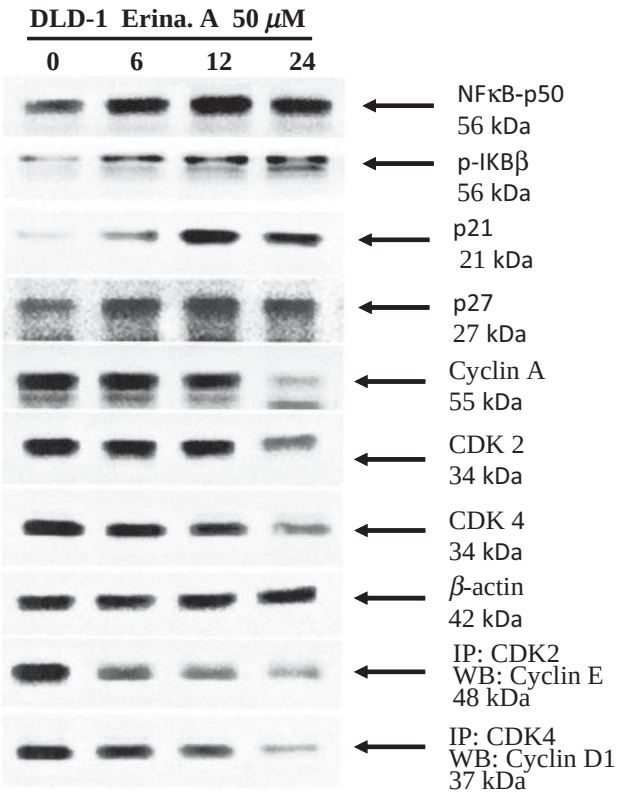


Fig. 5 – Effect of Erinacine A on the phosphorylation of IκBβ and p50 as well as expression of cell cycle associated proteins. DLD-1 cells were pretreated with Erinacine A for indicated time points. Whole cell lysate proteins were prepared and analysed by western blot, with β-actin serving as loading control. Erinacine A for the indicated times were analysed by 10% SDS-PAGE and subsequently treated with antibodies against IκBβ, p50, p21, p27, cyclin A, CDK2, and CDK4. The association of CDK2 with Cyclin E and CDK4 with Cyclin D1 were determined by immunoprecipitation followed by western blot with antibody.

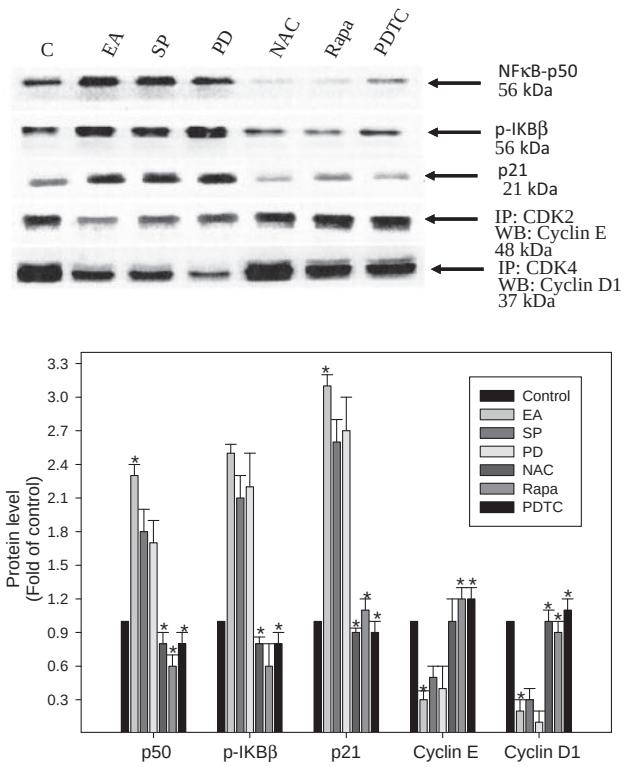


Fig. 6 – Effect of the kinase inhibitors in blocking erinacine A-induced cell cycle arrest. DLD-1 were incubated with various concentrations of the specific ERK1/2 inhibitor SP600125 (10 μM), the mTORp70 inhibitor rapamycin (50 nM), NFkB inhibitor PDTI or the ROS scavenger NAC (1 mM) for 1 h. Next, the cells were treated with erinacine A for 12 h. Total cell lysates of cells treated with or without erinacine A for the indicated time were extracted, and the expression of IκBβ, p50, p21, p27, and the association of CDK2 with Cyclin E and CDK4 with Cyclin D1 were determined, as described in "Materials and methods." Protein levels were quantified by densitometric analysis with the ratio of untreated control being set 1 fold. The quantitative data were presented as the mean of three repeats from one independent experiment. The data were presented as mean ± SD of three independent experiments. *$P < 0.05$, compared to the control group.

10 μM resulted in a significant antiproliferative effect against human colorectal cancer cells but not for normal HCoEpiC cells (Fig. 1). Furthermore, erinacine A treatment at the concentration of 10 μM for 24 h resulted in an induction of DLD-1 cell accumulation in the G1 phase (Fig. 2). We also found that erinacine A induced G1 arrest on other HCT-116 cells (unpublished data). It has been previously reported that a number of dried fruit body *Hericium erinaceus* extractions inhibited the MMP-2 and MMP-9 ability of human colon cancer and invasion via downregulation of the phosphorylation of ERK, JNK and p38 MAPK (Kim, Kang, Kim, Nam, & Friedman, 2011; Kim et al., 2013; Lee & Hong, 2010). Our study demonstrated that the cytotoxic effects of erinacine A on the CRC DLD-1 cell line appeared to be through G1 phase arrest and caused cell cycle arrest with the ROS production at a much earlier stage. This possibly played an important role in the erinacine A-triggered G1 phase arrest (Fig. 3). Figs. 4 and 5 show that treatment of DLD-1 cells with erinacine A decreased the level of phosphorylated ERK1/2 and p38 MAPK as a result of an upregulation of CDKI p21 and p27 at 12 and 24 h as well as a decrease of the CDK2/cyclin E and CDK4/cyclin D1 complexes as early as 6 and 12 h. Our results showed that erinacine A induced the cell cycle G1 arrest phase and affected G1 regulatory factors, resulting in an inhibition of the expression of cyclin A, CDK2 and CDK4 at 24 h in DLD-1 cells. Evidence indicates that the effect of cyclin A on the D1 cell G1 phase was mediated by the activation in the expression of JNK1/2 and p70S6K, and that upregulation of nuclear NFκB p50 and phosphor IKBβ was mediated through the differential regulation of the ROS generation and mTOR/p70S6K signalling pathways (Table 1). Erinacine A also markedly induced cell cycle arrest, whereas erinacine A treatment of HCoEpiC cells had no effect on growth.

Studies have shown that natural phytochemicals from certain plants have the capability to prevent cancer growth and

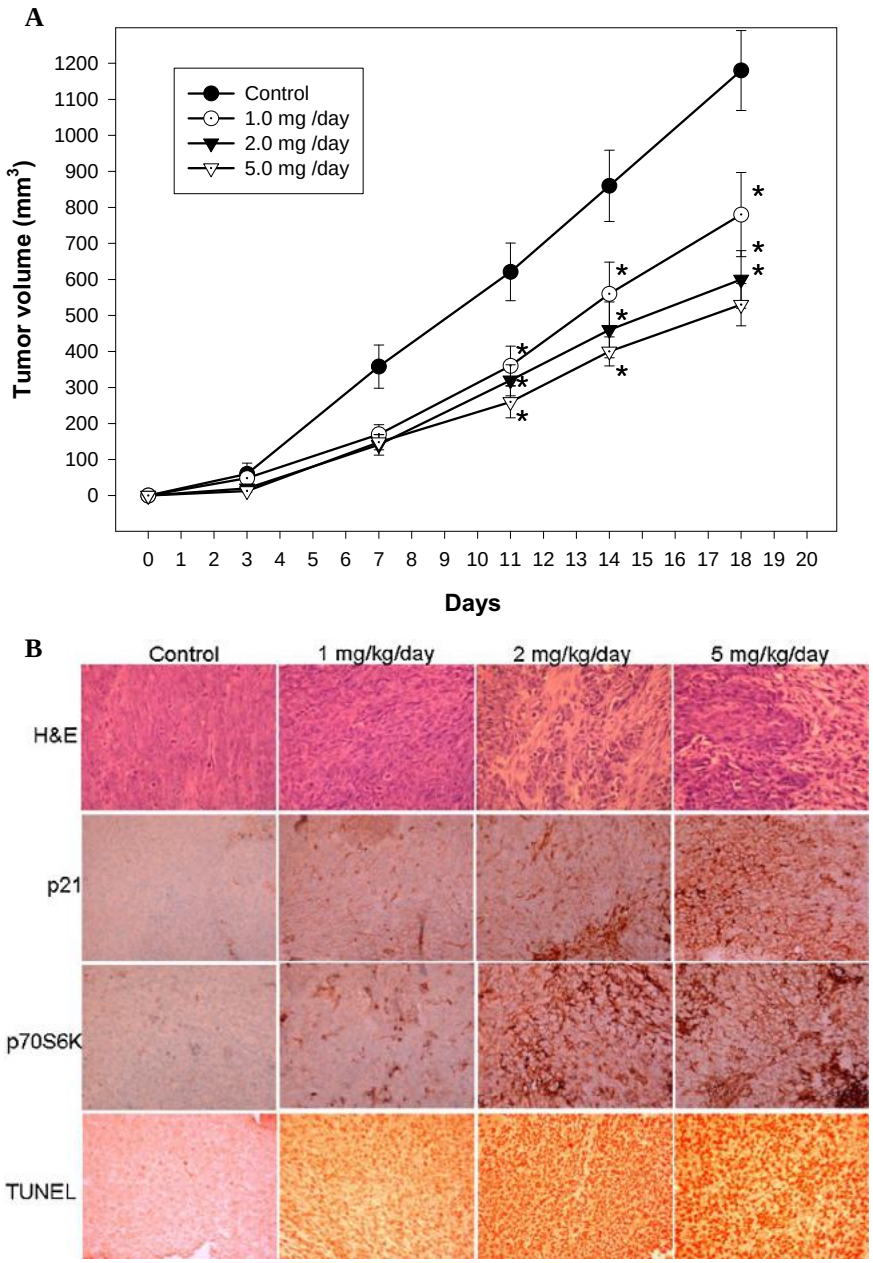


Fig. 7 – Growth inhibition of CRC xenograft by erinacine A. Nude mice were implanted subcutaneously with DLD-1 cells into flanks on day 0, then treated with or without (as a control) erinacine A as described in the "Materials and methods." (A) Time courses effect of erinacine A on the growth of DLD-1 xenograft was evaluated by determining the tumour volume every four days. (B) Immunohistochemical analysis of tumours was conducted, and multiple tumour fields were evaluated per group. Representative images for all groups from both experiments are presented. H&E staining revealed similar s.c. tumour morphology among all groups of tumours. P21 and p70S6K staining showed expressed tumour cells treated with erinacine A. TUNEL staining revealed significantly greater apoptosis in response to erinacine A in tumours.

that the produced elevated levels of ROS induce sustain MAPK activation, leading to cell death (Alexander et al., 2010; Ray et al., 2012). Furthermore, the MAPK p38 and JNK pathway has shown induction apoptosis in malignant cells in response to chemopreventive drugs, such as chrysin, berberine, and baicalein. This suggests that the MAPK pathway may be a promising target for anticancer effects (Hsu et al., 2007; Li-Weber, 2013; Zhu et al., 2007). Additionally, the PI3K/Akt/mTOR pathway is directly involved in inducing oxidative stress by phytochemicals, showing redox-sensitive activity-induced cell death (Ray et al., 2012; Teng et al., 2013). Previous data suggest that the potential antitumour activity of *H. erinaceus* mycelium hot water extracts may be mediated through a nuclear factor E2-related factor–antioxidant response element (Nrf2-ARE) pathway in human hepatocarcinoma cells (Jin et al., 2009). The aim of this study was to clarify the mechanisms of CRC growth inhibition and activation of the JNK1/2 MAPK and mTOR/p70S6K pathways, as well as the generation of ROS induced

by erinacine A in human DLD-1 cells. In addition, we suggest that these activation effects result from a downstream gene of p21 expression after erinacine A treatment and that these effects occur in response to ROS formation and mTOR/p70S6K phosphorylation via NF-κB activation (Fig. 6). Our results showed that the upregulation of p21 protein had an effect on the cell cycle arrest G1 of DLD-1 cells with a decrease in the association of the CDK2/cyclin E and CDK4/cyclin D1 complexes, which is preceded by the mTOR/p70S6K/NF-κB pathway, with an induction of ROS (Fig. 5).

With isolated erinacine A showing promising initial results, this study's findings are encouraging for the continued search for new dietary phytochemicals with effective chemoprevention. Many polyphenols and dietary compounds are effective chemopreventive agents, which have been explored to prevent tumour formation in animal experiments of anticancer processes (Lee et al., 2005). The water soluble components from *H. erinaceus* have been found to interfere with the MAPK phosphorylation of cancer cells, which induces apoptosis (Kim et al., 2011). The dried fruit body of *H. erinaceus* inhibits the metastasis of colon carcinoma cells to the lung cells (Kim et al., 2013). However, other antitumourigenic activities of *H. erinaceus* mycelium *in vivo* are not well-known. The present *in vivo* study demonstrated that intraperitoneal injections of erinacine A (1–5 mg/kg/day) treatment significantly reduced the volume and the number of mitotic cells and increased expression of p21 and p70S6K in DLD-1 CRC xenograft of nude mice (Fig. 7). Our results imply that the antiproliferation effect is one of the antitumourigenic mechanisms of erinacine A.

4. Conclusion

In this study, we evaluated that erinacine A inhibited the cell cycle G1 distribution as a result of the generation of the ROS and mTOR/p70S6K/NF-κB/p21 pathway (Fig. 8). *In vivo* study demonstrated that erinacine A decreased the growth of the xenografts of CRC cells in nude mice by inhibiting cell proliferation and apoptosis. These results provide insight into the antitumour effect of erinacine A of the *H. erinaceus* mycelium.

Conflict of interest

There is no financial/commercial conflict of interest.

Acknowledgments

Funding for this study was provided in part by research grants from the Tomorrow Medical Foundation, Taiwan. Funding for this study was provided in part by research grants, CLRPG8D0111, CMRPG6E0181, CMRPF6E0011, CMRPF6E0041, CMRPF6E0021, CMRPG8C1262, CMRPG6C0013, CMRPG6C0302, CMRPG6E0191, CMRPF6D0071 from Chang Gung Memorial Hospital-Kaohsiung Medical Center, Chang Gung Memorial Hospital, and Chang Gung University of Science and Technology, Chia-Yi Campus, Taiwan, and by the Ministry of Science and Technology, Taiwan (MOST 103-2622-B-255-001-CC3 and MOST 103-2313-B-255-001).

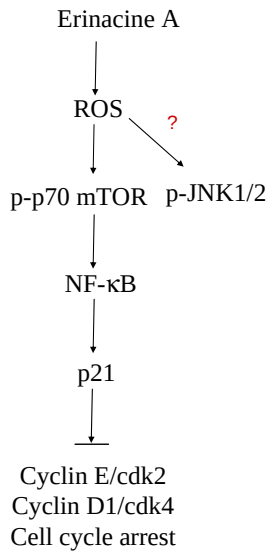


Fig. 8 – Schematic presentation of the signalling pathways involved in erinacine A induced cell cycle arrest in G1 phase of human DLD-1 cancer cells. The effect of erinacine A on the generation of ROS activates p70S6K, MAPK, and NF-kB pathways, which leads to p21 expression and cdk2/cyclin E and cdk4/cyclin D1 inactivation and increases cell cycle arrest in human DLD-1.

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